GE MEDICAL SYSTEMS

Technical Publications

Direction 46-305147
Revision 2

Signa® 1.5T
ATD-III

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, nature, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (141) 785-5052/8*323-5052 **immediately** after damage is found. At this time be ready to describe name of carrier, delivery date, consignee name, freight or express bill number, item damaged, and extent of damage.

Complete instructions regarding claim procedures are found in section “S” of the Policy & Procedure Bulletins.

3/12/92

GE MEDICAL SYSTEMS

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America

GE Medical Systems - Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON–RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇAO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
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注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

LIST OF EFFECTIVE PAGES

REVISION	DATE	PRIMARY REASON FOR CHANGE
REV 0	5/14/93	Initial Release
REV 1	6/10/93	Additional Information on Probe Interface Board
REV 2	2/97	Add Bloodborne Pathogen warning, Multilingual warning, update part numbers, reformat pages

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	2						
Damage	-						
Policy	0						
A	2						
B	Blank						
i	2						
ii	Blank						
1-1 - 1-4	2						
2-1 - 2-2	2						
3-1 - 3-3	2						
3-4	Blank						
4-1 - 4-7	2						
4-8	Blank						
5-1 - 5-2	2						
6-1 - 6-3	2						
6-4	Blank						
Data 1	2						
Data 2	Blank						
Data 3	2						
Data 4	Blank						
Data 5	2						
Data 6	Blank						
Blank							
GE Logo Pg.	-						

TABLE OF CONTENTS

SECTION 1 - INTRODUCTION

1-1 OVERVIEW

The system consists of a single-use-only, disposable, receive only coil, a Probe Interface Board installed within the Pelvic Array Coil, and the AutoTune Device III [ATD-III] which provide the interface, tuning, decoupling, and coil checking functions required to operate disposable coils with the GEMS Signa 1.5 Tesla MRI scanner system with the Phased Array Coil upgrade, and the Pelvic Array Coil.

The ATD III electronically optimizes the tuning of the disposable coil on a per-coil and per-patient basis by sweeping an electronically controlled reactance [varactor diode] through a range of values, and locking in the value which provides the optimum response at the nominal 63.86 MHz MRI system frequency. This allows the use of a coil which does not include internal tuning components. The components required to provide effective decoupling of the disposable coil from the RF excitation field during transmit are included in the ATD III Probe Interface Board, which is installed in the Phased Array Pelvic Coil.

For installation of the ATD-III Interface Board, refer to Section 6-2—*DECOUPLING DIODE REPLACEMENT*.

1-2 DESCRIPTION OF OPERATION OF THE ATD-III [ILLUSTRATION 1-1]

Coil Tuning/Decoupling Network

The network, located on the Probe Interface Board within the Pelvic Array Coil, consists of a parallel tuned circuit which includes a varactor diode. Tuning is achieved by stepping the voltage on the varactor diode. The ATD-III includes a matching network to transform the nominal coil impedance to 50 ohms, and a cable trap to provide cable current isolation for the disposable coil. A pair of crossed high speed switching diodes are connected across the coil input port. During transmit by the MR system, the RF voltage induced in the coil causes the diodes to conduct, placing a very low impedance across the coil output cable. Since the cable is one quarter wavelength long, at the coil end the cable appears as a very high impedance. This provides a blocking impedance in the coil loop to decouple the coil from the incident RF transmit field.

Interconnect Cable

A pair of cables within the Pelvic Array Coil output cable set provide a path for the RF signal output of the disposable coil to reach the ATD-III External Unit. The varactor tuning DC voltage is superimposed on one cable. The second cable provides a sensing signal for disposable coil connection detection.

63.86 MHz Signal Source

The ATD-III External Unit contains a crystal controlled RF source that applies a low level RF signal at the 63.86 MHz to the coil input network via the 50 ohm output. The RF source is only enabled during the automatic tuning function.

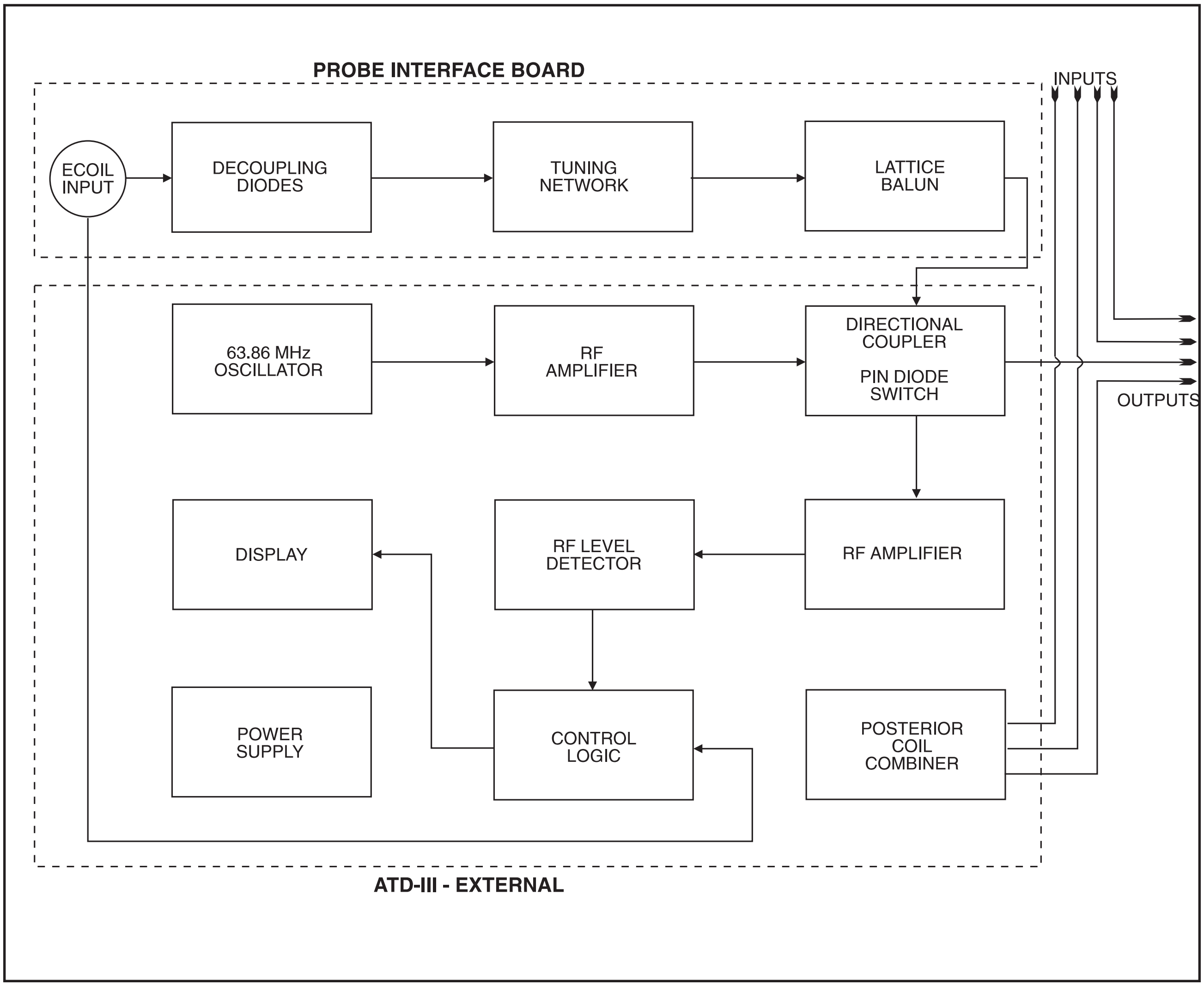
Description of Tuning Cycle

The ATD-III coordinates the operation of all the internal functions:

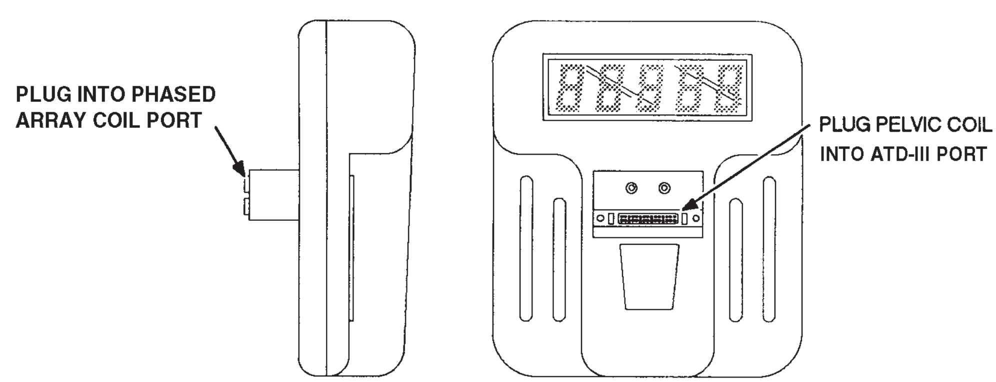
> The ATD-III, upon connection of the coil, begins ramping the tuning voltage applied to the varactor diode on the Internal Card through the range until a minimum return loss from the disposable coil is reached. The varactor voltage is locked at that point. A threshold comparator verifies that the coil is performing properly by checking that the return loss [reflected power] signal amplitude falls below the acceptance threshold.
>
> If no minimum is found in the return loss signal during tuning, or the return loss signal amplitude is above the acceptance threshold, the device displays a FAIL message to alert the operator that a coil failure has been detected.
>
> If an acceptable signal minimum is found, the ATD-III shuts off the RF signal source, switches the output of the Probe Interface Board to the sixth Multicoil input port, and displays a READY message, indicating that tuning has been successfully completed. The Endorectal Coil System is ready to operate in the Multicoil SCAN mode with the Pelvic Array Coil.

Display [ILLUSTRATION 1-2]

Prior to inserting the Endorectal Coil in the Endorectal Coil Port, the ATD-III will display the message **ECOIL**, indicating that the system is ready to have the Endorectal Coil connected to the port, and prompting the Operator to do so. During a normal coil tuning cycle, the display reads **TUNE** to alert the operator. The display switches automatically to either the **FAIL** or **READY** within 10 seconds of the beginning of the cycle. The display is steady state rather than clocked or multiplexed to ensure that no RF noise is generated during scanning.



BLOCK DIAGRAM OF ATD-III
ILLUSTRATION 1-1



ATD-III - FACE AND SIDE
ILLUSTRATION 1-2

1-2 DESCRIPTION OF OPERATION OF THE ATD-III - (cont'd)

Device Power

The **OFF**/**ON** functions of the ATD-III are controlled by the presence or absence of the ATD-III External Unit in the host system Multicoil Port, as the unit is powered by the Multicoil Port. When an Endorectal Coil is connected to the **PROBE INPUT** port of the Internal Card, the **TUNE** cycle is initiated. Removal of the coil connector will deactivate the ATD-III and **ECOIL** will be displayed. The status of the ATD-III is displayed at all times when connected.

SECTION 2 - SET UP AND CALIBRATION

2-1 SIGNA SCANNER VERIFICATION

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to “TLT PROCEDURE” calibration procedure located in Direction 15065, *Signa Troubleshooting Guide,* or Direction 15491, *Signa Advantage 1.5T & 0.5T System Troubleshooting Guide.*

The performance of the scanner with the Pelvic Array Coil, without the ATD-III, should be verified. The RF power level calibration from this test, is used to verify the ATD-III and disposable coil decoupling.

Select [**NEW STUDY**]—4.x, or [**New Exam**]—5.x/8.x for a new Landmark to be set. Place the Pelvic Array Coil on the patient table, and connect it to the Multicoil Port. Position the Pelvic Array coil, the TLT Test Phantoms, and the ATD-III Test Phantom—46-317824P1 without the Test Coil. Place the Anterior Coil of the Pelvic Array Coil over the top of the TLT Test Phantoms. See ILLUSTRATION 2-1.

LANDMARK the position, and advance the patient table to isocenter using the [**ADVANCE TO SCAN**] button.

Set up Scan Prescription as shown in ILLUSTRATION 3-1—4.x, or ILLUSTRATION 3-2—5.x/8.x.

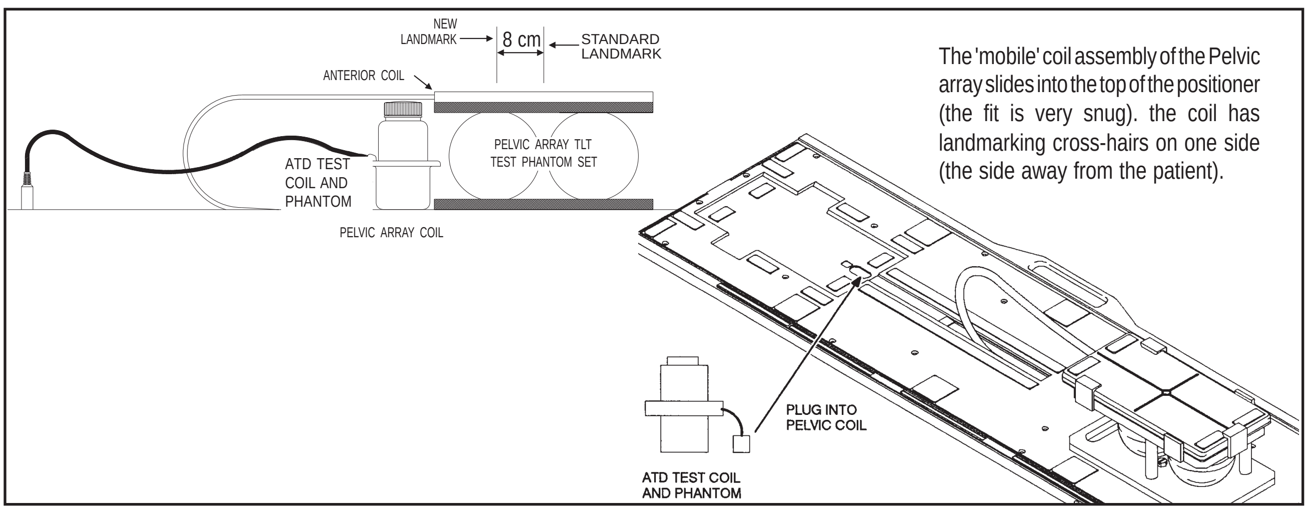
Select [**AUTOPRESCAN**] to properly calibrate the RF power level for the 90 degree and 180 degree pulses.

Select [**SCAN**]. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the resulting images. Record the Exam number and Series number for SNR calculations.

Select [**SCAN**] again. This second image will be used for determination of Pelvic Array mode SNR.

Determine the relative intensity of the image from each of the four coils in the Pelvic Array Coil to ensure that all four coils are contributing to the composite image equally.

Select [**CANCEL**]. Refer to Section 4 for SNR image analysis.



PHANTOM PLACEMENT
ILLUSTRATION 2-1

MAIN MENU [**NEW STUDY**]

PATIENT STUDY PARAMETERS
ID: **GESERVICE**
Patient Weight: **100 LBS**
Monitor SAR?: **N**
[**NEXT PAGE**]

PATIENT POSITION
Patient Entry: [**HEAD FIRST**]
Patient Position: [**SUPINE**]
Coil Type: [**OTHER COILS**]
[**PELVIC**]
Axial/Sag Landmark: [**STERNAL NOTCH**]
[**NEXT PAGE**]

IMAGING PARAMETERS
Image mode: [**SINGLE SCAN**]
Scan Mode: [**SAGITTAL**]
Pulse Sequence: [**MULTIPLE ECHO**]
Imaging Options: [**NONE**]
[**NEXT PAGE**]

SCAN TIMING
Number of Echos: [**1**]
Echo Time (TE): [**20 ms**]
Rep Time (TR): [**2000 ms**]
[**NEXT PAGE**]

SCANNING RANGE
Field of View: [**48 cm**]
Scan Thickness: [**3 mm**]
Scan Location: (S/I) **0**
FOV Center: (R/L **R0** (A/P) **A0**
[**NEXT PAGE**]

ACQUISITION TIME
Acq. Matrix: [**256 x 256**]
Imaging Time: [**1 NEX 8:58**]
Freq. Dir: [**R/L**]
Contrast: [**NO**]
Table Delta: **0**
[**NEXT PAGE**]

REVIEW
[**NEXT PAGE**]

SCAN OPERATIONS

PELVIC PHASED ARRAY SNR SCAN PROTOCOL—4.X SYSTEMS
ILLUSTRATION 2-2

PATIENT EXAM INFORMATION
ID: **GESERVICE**
Name:
Patient Weight: **300**
[**Patient Position**]

PATIENT POSITION
Patient Entry: [**Head First**]
Patient Position: [**Supine**]
Axial/Sag Landmark: [**Sternal Notch**]
Coil Type: [**Other**] [**Pelvic**]
Scan Plane: [**Sagittal**]
[**Imaging Parameters**]

IMAGING PARAMETERS
Image mode: [**2D**]
(SAR must be* ***ON*** **)* [**Monitor SAR**]
Pulse Sequence: [**Spin Echo]**
Imaging Options: [**None**]
or enter PSD Filename:
[**Next Screen**] or [**Scan Timing**]

SCAN TIMING
Number of Echos: [**1**]
Echo Time (TE): [**20 ms**]
Rep Time (TR): [**OTHER**] **2000 ms**
[**Scan Set—Up**] *(Release 5.3)*
[**Scanning Range**] *(Release 5.2)*

SCAN SET—UP *(Release 5.3)*
Auto CF: [**Peak**]
Receive Bandwidth: *for 0.5T only* [**16kHz**]
[**Scanning Range**]

SCANNING RANGE
Field of View: [**48**]
Slice Thickness: [**3 mm**]
Interscan Spacing: [**0**]
Start Location (I/S): **0**
End Location (I/S): **0**
No of Scan Locations **1**
FOV Center (L/R) **0** (P/A): **0**
[**<**] [**Acquisition Time**]

ACQUISITION TIME
Acq. Matrix: (freq.) [**256**]
Acq. Matrix: (phase) [**256**]
Frequency Direction: [**A/P**]
Phase FOV: default *(Release 5.3)*
Imaging Time: [**1 Nex**]
Contrast: [**NO**]
Table Delta **0**
[**Scan Ops**] *(Release 5.3)*
[**Scan Set—Up**] *(Release 5.2)*

SCAN SET—UP
Auto CF: [**Peak**]
Receive B/W (+/-): *for 0.5T ONLY* [**16 KHz**]
[**Scan Ops**]

SCAN OPERATIONS

PELVIC PHASED ARRAY SNR SCAN PROTOCOL—5.X/8.X SYSTEMS
ILLUSTRATION 2-3

SECTION 3 - FUNCTIONAL CHECKS

3-1 ATD-III PERFORMANCE VERIFICATION

Set up the Pelvic Array Coil, ATD-III, Test Phantom, Test Coil, and Test Cup and Phantoms on the patient transport table as in step 2-2. Use the same Landmark position. Do not connect the Test Phantom output cable to the Endorectal Coil Port on the Pelvic Array Coil at this time. Insert the ATD-III between the Pelvic Array Coil output cable and the Multicoil Port on the host system. Verify that the ATD-III displays the message **ECOIL** on the readout.

Insert the Test Coil Cable into the ENDORECTAL COIL INPUT of the Pelvic Array Coil. Observe that the display indicates **TUNE**, followed by **READY**. If this can not be achieved, go to Section 4.

Setup Scan Prescription as shown in ILLUSTRATION 3-1—4.x, or ILLUSTRATION 3-2—5.x/8.x.

Select [**Auto Prescan**] to properly calibrate the RF power level for the 90 degree and 180 degree pulses.

Select [**Scan**]. Observe the resulting images. See ILLUSTRATION 3-3. Ensure that there are no artifacts of any sort on any of the resulting images. Record the Exam number and Series number for SNR calculations.

CAUTION

If the ATD-III fails this test, do not use the device clinically. The Probe Interface Board in the Pelvic Array Coil must be replaced with a new board—see Section 6.

Select [**Scan**] again. This second image of the spheres will be used for determination of ATD-III SNR.

Inspect the images for visible ghosting [similar to motion artifacts] in the phase encode direction. If the images contain ghosting in the phase encode direction, an intermittent connection or component is suspected. Refer to Section 5 for surface coil troubleshooting.

Determine the relative intensity of the image from each of the four coils in the Pelvic Array Coil to ensure that all four coils are contributing to the composite image. The anterior coils may be slightly brighter because the two posterior coils of the pelvic array are combined to drive one port, slightly reducing the signal from each one. The signal from the Test Phantom and Test Coil should be brighter than the other pelvic array coils. If signal is absent from any of the coils, the ATD-III External Unit should be replaced, as it is not passing the signal from the pelvic array coils through to the Multicoil Port.

MAIN MENU [**NEW STUDY**]

PATIENT STUDY PARAMETERS
ID: **GESERVICE**
Patient Weight: **100 LBS**
Monitor SAR?: **N**
[**NEXT PAGE**]

PATIENT POSITION
Patient Entry: [**HEAD FIRST**]
Patient Position: [**SUPINE**]
Coil Type: [**OTHER COILS**]
[**PELVIC**]
Axial/Sag Landmark: [**STERNAL NOTCH**]
[**NEXT PAGE**]

IMAGING PARAMETERS
Image mode: [**SINGLE SCAN**]
Scan Mode: [**SAGITTAL**]
Pulse Sequence: [**MULTIPLE ECHO**]
Imaging Options: [**NONE**]
[**NEXT PAGE**]

SCAN TIMING
Number of Echos: [**1**]
Echo Time (TE): [**20 ms**]
Rep Time (TR): [**2000 ms**]
[**NEXT PAGE**]

SCANNING RANGE
Field of View: [**48 cm**]
Scan Thickness: [**3 mm**]
Scan Location: (S/I) **0**
FOV Center: (R/L **R0** (A/P) **A0**
[**NEXT PAGE**]

ACQUISITION TIME
Acq. Matrix: [**256 x 256**]
Imaging Time: [**1 NEX 8:58**]
Freq. Dir: [**R/L**]
Contrast: [**NO**]
Table Delta: **0**
[**NEXT PAGE**]

REVIEW
[**NEXT PAGE**]

SCAN OPERATIONS

ATD-III SNR SCAN PROTOCOL—4.X SYSTEMS
ILLUSTRATION 3-1

PATIENT EXAM INFORMATION
ID: **GESERVICE**
Name:
Patient Weight: **300**
[**Patient Position**]

PATIENT POSITION
Patient Entry: [**Head First**]
Patient Position: [**Supine**]
Axial/Sag Landmark: [**Sternal Notch**]
Coil Type: [**Other**] [**Pelvic**]
Scan Plane: [**Sagittal**]
[**Imaging Parameters**]

IMAGING PARAMETERS
Image mode: [**2D**]
(SAR must be **ON** *)* [**Monitor SAR**]
Pulse Sequence: [**Spin Echo]**
Imaging Options: [**None**]
or enter PSD Filename:
[**Next Screen**] or [**Scan Timing**]

SCAN TIMING
Number of Echos: [**1**]
Echo Time (TE): [**20 ms**]
Rep Time (TR): [**OTHER**] **2000 ms**
[**Scan Set—Up**] *(Release 5.3)*
[**Scanning Range**] *(Release 5.2)*

SCAN SET—UP *(Release 5.3)*
Auto CF: [**Peak**]
Receive Bandwidth: *for 0.5T only* [**16kHz**]
[**Scanning Range**]

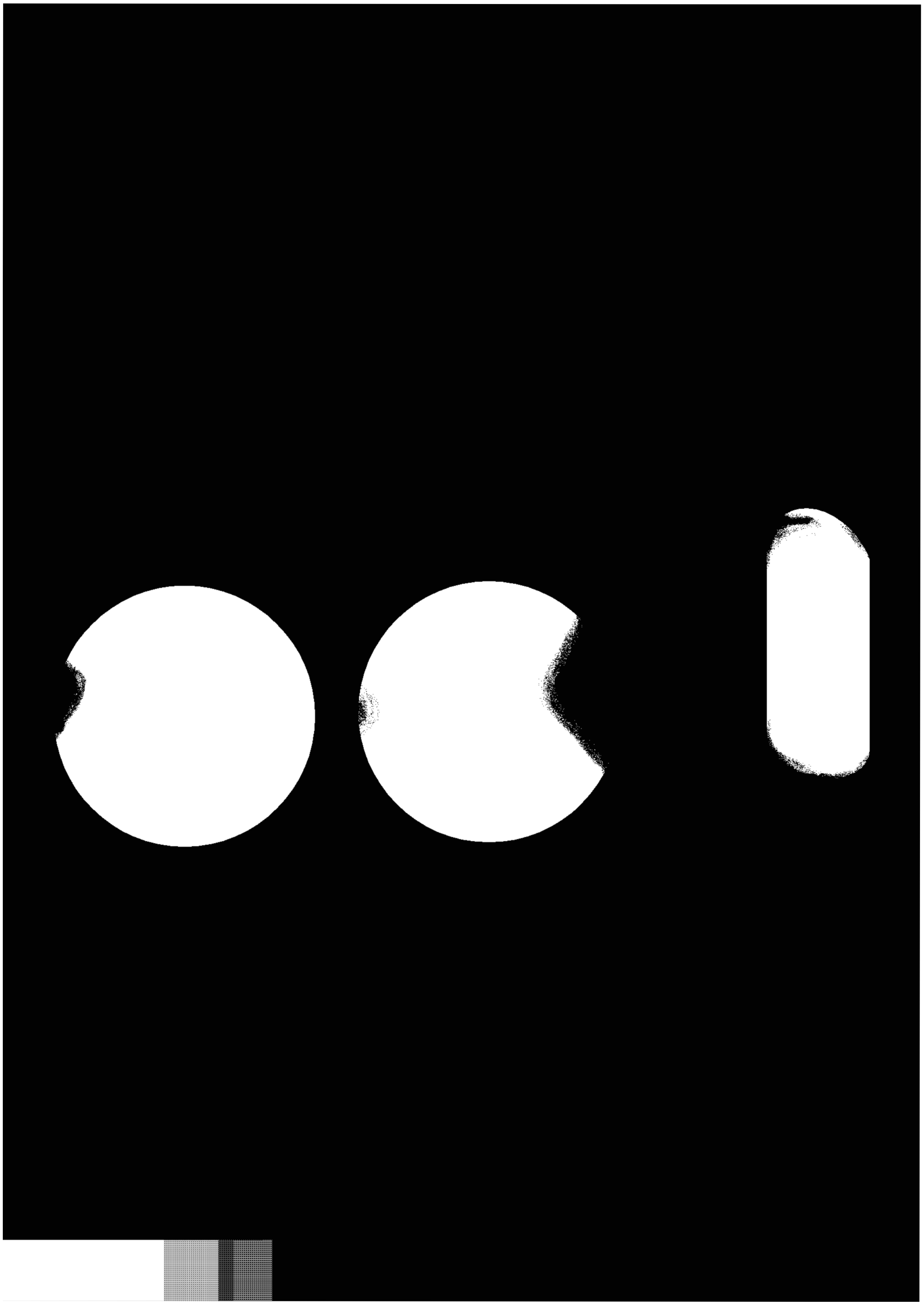
SCANNING RANGE
Field of View: [**48**]
Slice Thickness: [**3 mm**]
Interscan Spacing: [**0**]
Start Location (I/S): **0**
End Location (I/S): **0**
No of Scan Locations **1**
FOV Center (L/R) **0** (P/A): **0**
[**<**] [**Acquisition Time**]

ACQUISITION TIME
Acq. Matrix: (freq.) [**256**]
Acq. Matrix: (phase) [**256**]
Frequency Direction: [**A/P**]
Phase FOV: default *(Release 5.3)*
Imaging Time: [**1 Nex**]
Contrast: [**NO**]
Table Delta **0**
[**Scan Ops**] *(Release 5.3)*
[**Scan Set—Up**] *(Release 5.2)*

SCAN SET—UP
Auto CF: [**Peak**]
Receive B/W (+/-): *for 0.5T ONLY* [**16 KHz**]
[**Scan Ops**]

SCAN OPERATIONS

ATD-III SNR SCAN PROTOCOL—5.X/8.X SYSTEMS
ILLUSTRATION 3-2



PELVIC PHASED ARRAY COIL WITH ATD-III IMAGE
ILLUSTRATION 3-3

SECTION 4 - SNR IMAGE ANALYSIS

4-1 SNR IMAGE ANALYSIS FOR 4.X SYSTEMS

On the left touch screen, touch the **UTILITIES** soft key. When the next screen appears, touch the **CLIPS** key.

Type **1** and **ENTER** on the next screen to run CLIPS.

Select the image processor at the Operator Console by typing **1** and **ENTER** at the next screen. It is not necessary to boot the image processor, so select **N** and **ENTER** when asked at the next screen.

It will take several seconds for the **CLIPS>** prompt to appear. When it does, type **LIST(STUDY);** and **ENTER**. A screen listing the available studies will appear. Type **S** and **ENTER** to enable the **SELECT STUDY** function. Enter the study number of the Body Coil SNR scans.

At the **CLIPS>** prompt, type **LIST(SERIES);** and **ENTER**. A list of the available series will appear. Type **S** and **ENTER** to select the proper series. Generally, the desired series will be the first one. Type "**1**" and **ENTER** to select the series.

To view several images at one time, type **VIEW(2,2)**; at the **CLIPS>** prompt. This will display images in a pattern of four images; the upper left image will be #1, the upper right image will be #2, the lower left will be #3, and the lower right will be #4. No images are displayed at this time.

At the **CLIPS>** prompt, type **LIST(IMAGE);** and **ENTER**. Type **2** and **ENTER** to select image by number. Type **A** to select all images. Type **E** to exit the image selection.

It is necessary to load the images to be analyzed to the **DISPLAY** function of CLIPS. Type **DIS(ALL);** and **ENTER**. Two images should now be displayed, both in the top row.

To obtain a mean signal value for the signal to noise ratio calculation, type **ROI;** at the **CLIPS>** prompt. A menu will appear allowing a selection of cursor types. Choose the elliptical cursor by typing **2** and **ENTER**.

Adjust the size of the cursor using the [**CURSOR POSITION/SIZE**] keys on the console and the trackball to an area covering approximately 80% of the phantom image in the upper left corner [**Display 1**]. Adjust the position of the cursor circle to center it on the phantom. Press the **ENTER** key on the keyboard to allow CLIPS to calculate the ROI parameters. Record the **MEAN SIGNAL** value for later use in the signal to noise ratio calculation.

Type **#3=#1-#2;** at the **CLIPS>** prompt. This will subtract the two images, creating a noise image in **Display 3**.

Type **#1=#3;** at the CLIPS> prompt. This action moves the image to the upper left display window to allow ROI calculations to be made.

4-1 SNR IMAGE ANALYSIS FOR 4.X SYSTEMS - (cont'd)

At the **CLIPS>** prompt, type **ROI;** to select the image analysis function. Type **2** and select an elliptical cursor. Size and position the cursor to the same conditions as in step 4-1-10. Press the **ENTER** key, and record the **Standard Deviation** value for later use in the signal to noise ratio calculation.

Use the following formula to calculate the signal to noise ratio:

SIGNAL TO NOISE RATIO = 1.414 * MEAN SIGNAL (from 4-1-10) / SD (from step 4-1-13)

Record the value determined for the signal to noise ratio and the coil type [surface or body] of the measurement on the *SNR QA DATA TABLE*.

Repeat steps 4-1-5 to 4-1-14 using the surface coil images to determine the signal to noise ratio for the surface coil scans.

Determine the relative performance of the surface coil against the body coil reference by dividing the surface coil signal to noise ratio value by the body coil signal to noise ratio value.

To end the CLIPS utility when all image calculations are completed, type **BYE;** and **ENTER** at the **CLIPS>** prompt.

Select the exit function by typing **5** at the select function screen. Touch the **CANCEL** soft key on the screen to exit the **UTILITIES** functions and return to normal system operation.

Upon exiting **CLIPS**, reset the **AUTO CENTER FREQUENCY** mode by touching the **SCAN MODES** key, and then **DEFAULT AUTO CF** followed by the **EXECUTE** key.

Record the date and value calculated in the appropriate column under “SNR Data QA Check” of the Data Table.

4-2 SNR IMAGE ANALYSIS FOR 5.X SYSTEMS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

Touch [**UTILITIES**], [**MRTools**], [**Image Quality**], then [**SNRTest**]. The SNR Test screen is displayed. See ILLUSTRATION 5-1.

Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select [**List Exams**], [**List Series**], or [**List Images**] to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

4-2 SNR IMAGE ANALYSIS FOR 5.X SYSTEMS - (cont'd)

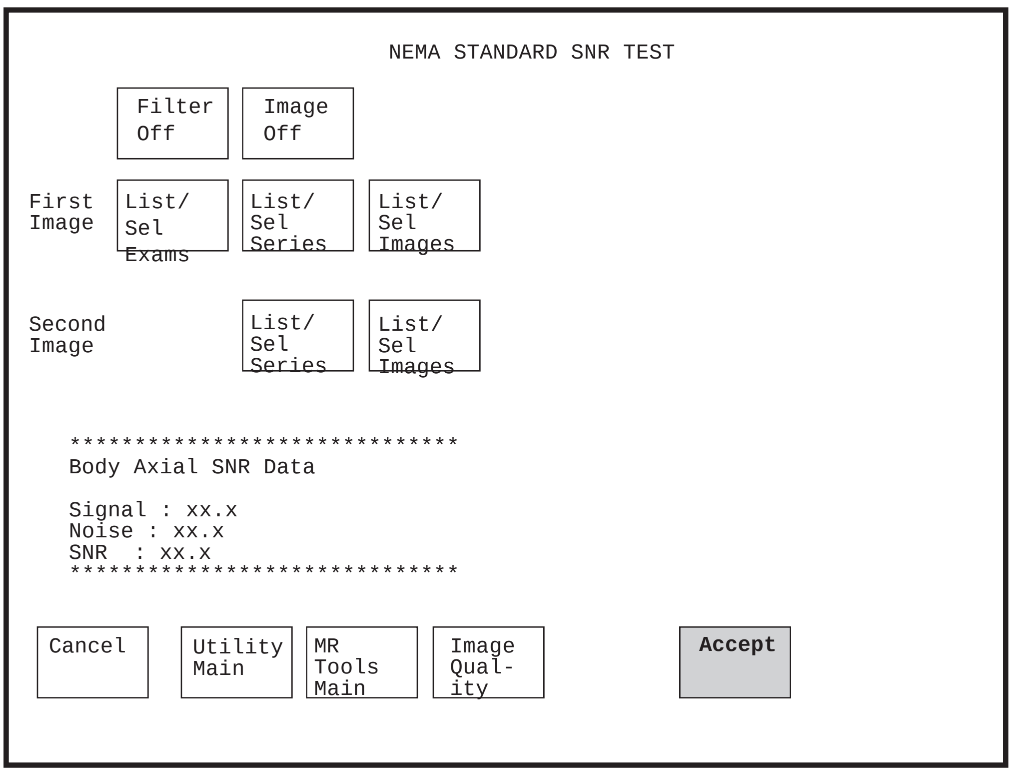
Enter second image series and image numbers. If series or image numbers are not known, select [**List Series**] or [**List Images**] to display list. The second image Exam Number is defaulted to the same as first image.

Select [**Filter Off**] and [**Image Off**].

Touch [**Accept**] to begin analysis. The final values are displayed on the screen. See ILLUSTRATION 4-1.

Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table.

Touch [**Continue**] then select the next exam and repeat the above analysis for each image pair



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 4-1

4-3 SNR IMAGE ANALYSIS FOR 8.X SYSTEMS

Description

The SNR Tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported.

Procedure

Select [**Service Desktop**], [**Calibration/Checks**], then [**Image Quality**].

<<< NEMA Image Quality Analysis >>>

1. Signal To Noise Check
2. Slice Offset Checks
3. Slice Thickness/Resolution Check
4. T2 Uniformity Check
5. Full Field Distortion Check
6. Exit NEMA test

Select Test: ....**1 [ENTER]**

```
**************************
*  Select First Image    *
**************************
```

```
========================
 Image Selection Menu
========================
```

Current Selection:

Exam_No = 50210, Series_No = 1, Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test
YOUR CHOICE :...**A [ENTER]** *(Enter appropriate selection, A-F to choose image.)*

Exam No. ?...**50201 [ENTER] (***Enter appropriate number.)*

(Iterate image selection process until 'Current Selection' *is correct.)*

4-3 SNR IMAGE ANALYSIS FOR 8.X SYSTEMS - (cont'd)

```
     =========================
      Image Selection Menu
     =========================

Current Selection:

    Exam_No = 50201,   Series_No = 1,  Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test
YOUR CHOICE :
Invalid Entry
```

('Invalid Entry', *always occurs here.*)

```
     =========================
      Image Selection Menu
     =========================

Current Selection:

    Exam_No = 50201,   Series_No = 2,  Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test
```

YOUR CHOICE :..**X [ENTER]** (*Perform this step when* 'Current Selection' *is correct.*)

```
***************************
*  Select Second Image   *
***************************
```

4-3 SNR IMAGE ANALYSIS FOR 8.X SYSTEMS - (cont'd)

```
=========================
 Image Selection Menu
=========================
```

Current Selection:

Exam_No = 50210, Series_No = 1, Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test
YOUR CHOICE :..**A [ENTER]** *(Enter appropriate selection, A-F to choose image.)*

Exam No. ?..**50201 [ENTER]** (*Enter appropriate number.)*

(Iterate image selection process until 'Current Selection' *is correct.)*

```
=========================
 Image Selection Menu
=========================
```

Current Selection:

Exam_No = 50201, Series_No = 2, Image_No = 2

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test
YOUR CHOICE :..**X [ENTER]** (*Perform this step when* 'Current Selection' *is correct.*)

4-3 SNR IMAGE ANALYSIS FOR 8.X SYSTEMS - (cont'd)

```
************************************
*       SNR results              *
*                             *
* Signal = 801.040222  Noise = 10.062437  snr = 79.606979 *
************************************
```

```
<<< NEMA Image Quality Analysis >>>

        1. Signal To Noise Check
        2. Slice Offset Checks
        3. Slice Thickness/Resolution Check
        4. T2 Uniformity Check
        5. Full Field Distortion Check
        6. Exit NEMA test
```

Select Test:.. **6 [ENTER]**

Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table.

SECTION 5 - ATD-III TEST AND TROUBLESHOOTING PROCEDURES

5-1 TUNING TEST

1 Place the Pelvic Array Coil on the patient table. Insert the ATD-III into the Multicoil Port on the system. Insert the Pelvic Array Coil connector into the front connector of the ATD-III. Verify that the ATD-III displays the **ECOIL** message.

2 Position the Test Coil and Phantom on a support of two comfort pads in a stack on the Pelvic Array Coil.

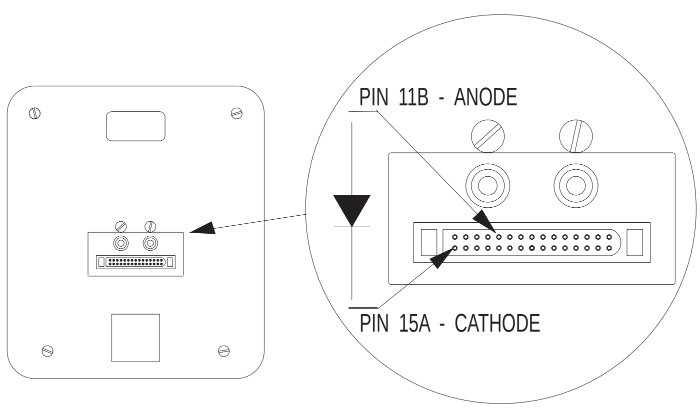
3 Insert the connector of the Test Coil into the ENDORECTAL COIL INPUT JACK on the Pelvic Array Coil. Observe the Readout Display; the**TUNE** message should be visible for a period not exceeding ten seconds, followed by **READY**; indicating that the ATD-III has successfully tuned the coil.

5-2 PIN DIODE TEST

1 Select the DIODE TEST function on the Digital Multimeter.

2 Connect the NEGATIVE lead of the DMM to Pin 15A of the 30 pin Bendix connector on the rear of the ATD-III. Connect the POSITIVE lead to Pin 11B of the 30 pin Bendix connector on the rear of the ATD-III. See ILLUSTRATION 5-1.

3 A reading of .400 to .900 should be observed on the DMM.



PIN DIODE TEST
ILLUSTRATION 5-1

5-2 PIN DIODE TEST - (cont'd)

4 Reverse the lead connections in step 2, connecting the POSITIVE lead of the DMM to Pin 15A and the NEGATIVE lead to Pin 11B.

5 A reading of INFINITY should be observed on the DMM.

6 If a reading below .400 is observed in either direction the PIN diode in the ATD-III is shorted.

7 If a reading of above .900 is observed in step 4, the PIN diode is defective.

8 If a reading of INFINITY is observed in both directions the PIN diode in the ATD-III is open, or the combiner card is defective.

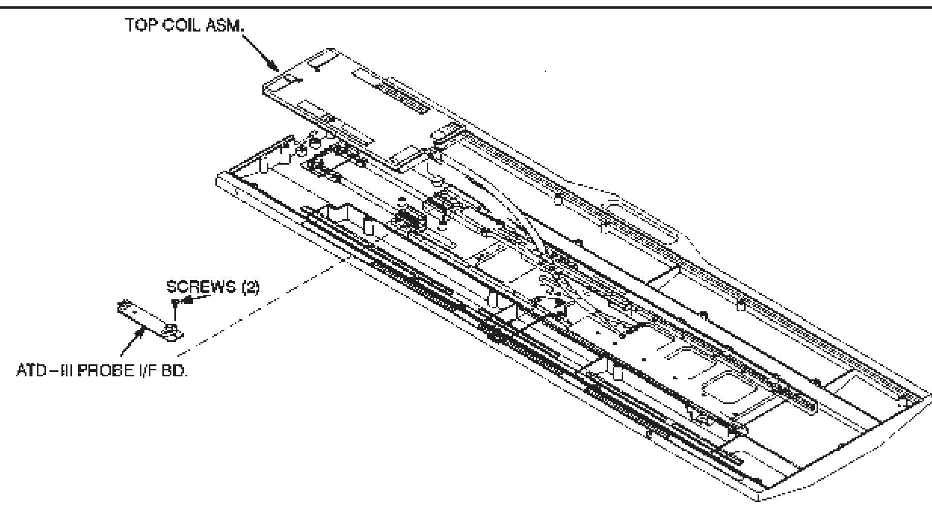
SECTION 6 - ATD-III SERVICE PROCEDURES

WARNING

BIO HAZARD! EQUIPMENT BEING RETURNED FROM USE IN A CLINICAL SETTING MUST BE CLEAN AND FREE OF BLOOD AND OTHER INFECTIOUS SUBSTANCES. THE DEPARTMENT OF TRANSPORTATION (DOT) HAS RULED THAT ITEMS THAT WERE SATURATED AND/OR DRIPPING WITH HUMAN BLOOD THAT ARE NOW CAKED WITH DRIED BLOOD; OR WHICH WERE USED OR INTENDED FOR USE IN PATIENT CARE ARE REGULATED MEDICAL WASTE FOR TRANSPORTATION PURPOSES AND MUST BE TRANSPORTED AS A HAZARDOUS MATERIAL. UNDER NO CIRCUMSTANCES SHOULD A PART OR EQUIPMENT WITH VISIBLE BODY FLUIDS BE TAKEN OR SHIPPED FROM A CLINIC OR SITE (FOR EXAMPLE, SURFACE COILS).

Employees shall follow proper decontamination procedures for clean up of bloodborne pathogens. Clean the ATD-III with a soft cloth dampened with a 10% bleach-in-water solution. Do not allow any of the solution to enter the ATD-III. It is the responsibility of the GEMS employee to insure the part/equipment has been properly decontaminated prior to shipment.

Part Number	Device	Quantity
46-328534G2	ATD-III with Probe Interface Board	1



REPLACEMENT OF ATD-III PROBE INTERFACE CIRCUIT BOARD
ILLUSTRATION 6-1

6-1 PIN DIODE REPLACEMENT

As ATD-III calibration and operation may be affected by PIN diode replacement, field replacement of the RF protection diode is not recommended.

6-2 DECOUPLING DIODE REPLACEMENT—ENTIRE PROBE INTERFACE BOARD MUST BE REPLACED

Replacement Phased Pelvic Array Coils contain a dummy Probe Interface Board. It is necessary to remove the Probe Interface Board from the defective Phased Pelvic Array Coil and install it in the replacement Phased Pelvic Array Coil. The dummy Probe Interface Board can be saved or disposed.

Disassemble the top and bottom housings. See Section 6-3.

Disconnect the two SMB connectors from the Dummy Probe or ATD-III Probe Interface Board (46-328131P2). See Illustration 6-1.

Remove the two screws holding the Dummy Probe or ATD-III Probe Interface Board in place. Retain screws.

Remove the defective ATD-III Probe Interface Board.

Install the new ATD-III Probe Interface Board in place. See Illustration 6-1.

Install the screws retained from Step 3 on the ATD-III Probe Interface Board.

Reconnect the two SMB connectors to the ATD-III Probe Interface Board by attaching cable plug P-3 to SMB connector J-3 and cable plug P-7 to SMB connector J-7.

Assemble the top and bottom housings together. Refer to Section 6-3.

Note

All antenna loop elements are tuned at the factory to provide the proper input impedance. Since antenna elements 1 and 2 are connected to elements 3 and 4, the top and bottom Pelvic Array circuit boards are field replaceable as a unit. These circuits must remain together as a matched set.

6-3 DISASSEMBLY/ASSEMBLY OF PELVIC PHASED ARRAY COIL

Remove pad from top side of Pelvic Coil assembly. See Illustration 6-2.

Remove the eighteen screws holding the top and bottom housing together.

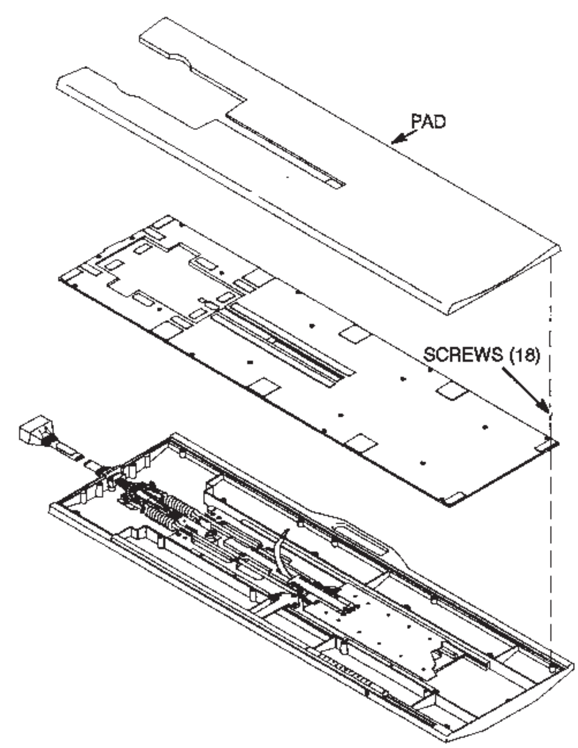
Separate the two housings from each other.

To reassemble, put the two housings together and **lightly** tighten the eighteen screws.

Note
The screws are very susceptible to breakage from overtightening.

Attach pad to top side of Pelvic Coil assembly.

Proceed with Performance Verification. See Section 3.



ASSEMBLY/DISASSEMBLY OF PELVIC PHASED ARRAY COIL
ILLUSTRATION 6-2

SNR QA DATA TABLE

Use the space provided below to record the calculated signal to noise ratio—SNR, data obtained from Section 4, *SNR IMAGE ANALYSIS.* After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2 for Phased Array Coil or column 3 for ATD-III coils, enter the date in column 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: ______________________

Phased Array Coil Value: ____________________

ATD-III Coil Value: _________________________

Date: ____________

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1 Date	2 Phased Array Coil QA SNR Value:	3 ATD-III Coil QA SNR Value:	4 Are new values divided by original values > 85%	
				Y/N
				Y/N
				Y/N
				Y/N
				Y/N
				Y/N
				Y/N
				Y/N
				Y/N
				Y/N
				Y/N

Make additional copies of this document as needed.

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

To allow for proper assessment of defective returned coils, both sides of this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: ______________________________

SITE NAME: ______________________________

SITE ADDRESS: ______________________________

____

SERVICE ENGINEER: ______________________________

ATD SERIAL NUMBER: ______________________________

DATE ATD INSTALLED: ______________________________

DESCRIPTION OF ATD PROBLEM:

DESCRIPTION OF PROBE INTERFACE BOARD PROBLEM:

TLT PROCEDURE RECORDING FORM

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to “TLT PROCEDURE” calibration procedure located in Direction 15065, *Signa Troubleshooting Guide,* or Direction 15491, *Signa Advantage 1.5T & 0.5T System Troubleshooting Guide.* Use the table to record the TLT results of the defective coil.

SITE:		**NAME:**	**DATE:**
TLT FILE NUMBER:			
SNR:	NOISE (0)		
	NOISE (1)		
	NOISE (2)		
	NOISE (3)		
	NOISE (C)		
	SNR MEAN (0)		
	SNR MEAN (1)		
	SNR MEAN (2)		
	SNR MEAN (3)		
	SNR MEAN (C)		
TR MAP: (Combined Results)	89–91 (C)	%	%
	85–95 (C)	%	%
	65–115 (C)	%	%
	FLIP MEAN (C)		
	FLIP SDV (C)		
	RCV MEAN (C)		
	RCV SDV (C)		
	DIFF MEAN (C)		
	DIFF SDV (C)		

GE MEDICAL SYSTEMS

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America

GE Medical Systems - Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

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